Project Executable Files

Date	2 july 2026
Team ID	xxxxxx
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

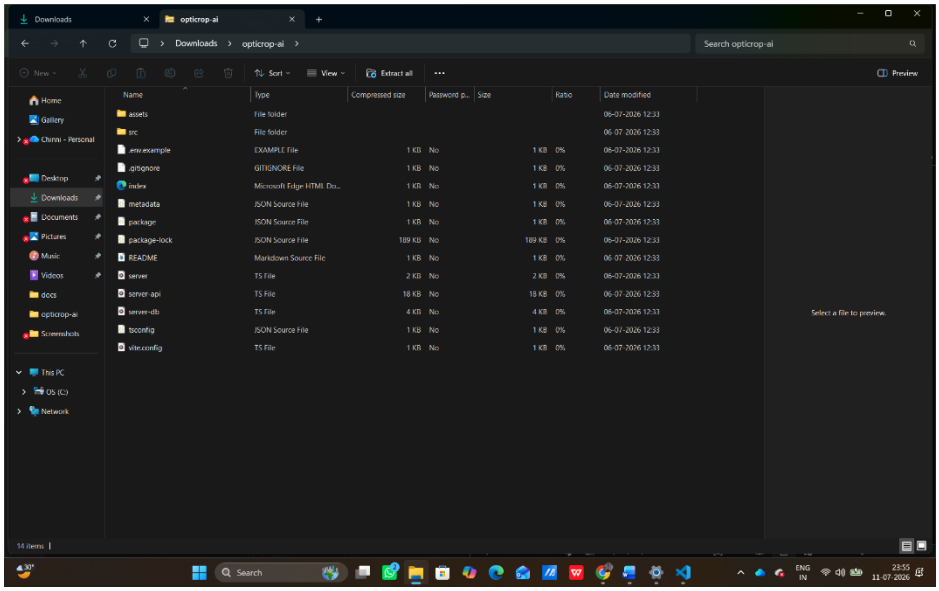
Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	yes
9	Test files and test results	yes

10	Demo video or walkthrough (if required)	yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-ai-463011982132.asia-southeast1.run.app
Login Credentials (Demo)	Login credentials are user name , password
Platform / Hosting Provider	Render.
Repository Link	https://github.com/Sumanthkalyan-07/opticrop-ai
Demo Video Link	https://drive.google.com/file/d/1bkAYkAVqGf1d1aYtVBLvpMyLFvdPyjMH/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

The project was developed using a set of powerful Python-based tools and libraries for data processing, machine learning, visualization, and deployment. These technologies provide the foundation for efficient model development, analysis, and application deployment.

Anaconda Navigator: A desktop graphical user interface used to launch applications and manage Python packages and environments for data science. https://www.anaconda.com/download

PyCharm: An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools. https://www.jetbrains.com/pycharm/

NumPy: Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. https://numpy.org/doc/stable/

Pandas: Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series. https://pandas.pydata.org/docs/

Scikit-learn: Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. https://scikit-learn.org/stable/

Matplotlib: Comprehensive library for creating static, animated, and interactive visualizations in Python. https://matplotlib.org/stable/

Seaborn: Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics. https://seaborn.pydata.org/

Flask: A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. https://flask.palletsprojects.com/

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	**1. Architectural & State Limitations** • **Stateless Operation:** In its current production form, the application does not persist data. If a user refreshes the page, their entire generation history, selected inputs, and custom tailored copy are permanently lost. • **Lack of User Authentication:** The system lacks multi-tenant isolation. Without a database tier containing a Users table, it is impossible to implement personalized user dashboards, save favorite captions, or limit API usage per user.	When a system runs slow, crashes, or cannot handle data efficiently, the limitations are rectified through architectural changes: • **Vertical Scaling (Upgrading):** Adding more power (CPU, RAM, SSD) to a single machine to handle heavier workloads. • **Horizontal Scaling (Outgrowing):** Adding more machines to share the load, often managed by load balancers.
2	**. API & Performance Constraints** • **Groq API Rate Limits:** The backend relies entirely on a single API token for the Groq Cloud SDK. High concurrent traffic from multiple users can trigger rate limits (429 Too Many Requests), causing the application to stall. • **Single-Point-of-Failure Dependency:** Because the application does not cache previous generations or fall back to alternative LLM providers, any operational	**2. Machine Learning & Data Limitations** If a machine learning model (like the OptiCrop engine) is inaccurate or biased, the limitations are rectified during the data and training phases: • **Overfitting:** When a model memorizes training data but fails on new data. Rectified by

	downtime on Groq's API directly results in a total outage of the caption generation service.	**Regularization** (penalizing complexity), adding **Dropout layers**, or gathering **More diverse training data**.
3	**Data & Scalability Constraints** • **Stringent Input Lengths:** The current UI and prompt structure limit user input descriptions. Extremely detailed product descriptions or nuanced brand guidelines may get truncated, resulting in less accurate outputs. • **Normalized Data Overhead:** While the proposed relational schema ensures data integrity, breaking down a single generation event into three separate tables (Users, Generations, Generated_Outputs) requires complex JOIN queries. As the historical log scales into millions of rows, this can introduce read latency without proper indexing.	No matter the project, teams generally rectify limitations by moving through these four distinct phases: **1.Identification & Logging:**Phase 1. Detecting the failure point through stress testing, user feedback, or error logs. The limitation must be clearly defined (e.g., "The system crashes when 1,000 users input soil data simultaneously").